- **Lines**
  (a) $aRe(z) + bIm(z) = c$
  (b) $|z - z_1| = |z - z_2|$

1. Plot the line $\frac{|z-3|}{|z-2+i|} = 1$

2. Plot the region $3Re(z - 2 + 3i) \leq 1 - Im(z - 1)$

- **Rays**
  (a) Defined via the arg function, usually of form $Arg(z - z_1) = \theta$
  (b) Note $Arg(0)$ is not defined, so there is an open circle at the terminus

1. Interpret what a does in the graph of $Arg(az) = \frac{\pi}{4}$

2. Interpret what b does in the graph of $Arg(z - b) = \frac{\pi}{4}$

3. Find the rule for the graph of $Arg(iz - 1) = \frac{\pi}{3}$ in Cartesian form

4. Find the rule for the graph of $Arg(z - 1) = Arg(z + 1)$ in Cartesian form

- **Circles**
  (a) $|z - z_1| = a$
  (b) $(z - z_1)(\bar{z} - \bar{z}_1) = a$ (This is similar to form (a), how?)
  (c) $|z - z_1| = b|z - z_2|$, $b > 0, b \neq 1$

  1. $4(2z + 3 - 2i)(2\bar{z} + 3 + 2i) = 25$
  2. $|z - 2| = 2|z + 1|$

1. Find the cartesian form of the graph of $|z + 1 - 2i| = 2$

2. Find the cartesian form of the graph of $Arg(z + i) = \alpha, \alpha \in (0, \frac{\pi}{2})$

3. Hence find the simplest form of the exact minimum value of $Arg(z+i)$ where z satisfies $|z+1-2i| = 2$

4. Prove $|z + 3 - i|^2 = 4|z - 1 + 3i|^2$ if and only if $|3z - 7 + 13i|^2 = 128$

   (a) State the radius and centre of the circle described above

(b) Find the value z on the circle with the largest magnitude